My work is unified by an interest in developing empirical models that accurately represent the complexity in real-world social and political processes. I have ongoing collaborative research agendas in research methods, international relations, American politics and government communication processes. In research methods, my primary research interests are in modeling networks and collective decision-making processes. I also have a methodological research agenda in robust estimation and out-of-sample model comparison. Within international relations, I study interstate relational event modeling and forecasting. Within American politics, I study judicial politics and legislative collaboration. Also, I have recently begun a major project on communication processes within government organizations. Below I describe the major themes in my research in greater detail and characterize the relationships among them.

The choices and/or actions of one social entity often depend upon the choices and/or actions of others. This interdependence is fundamental to many theoretical accounts of political decision-making and interaction. For example, the chief justice of the U.S. Supreme Court is thought to vote with the majority on the merits vote so as to assign opinion authorship. In the international arena, the number of signatories to a defense pact increases the protective utility of the pact derived by each state involved. Due to the dominance of the regression paradigm in empirical social science research, interdependence is often ignored insofar as it cannot be represented by exogenous covariates. I have an ongoing research agenda, which began with my dissertation, on the development and application of statistical methods that can precisely represent and test hypotheses regarding theories of interdependence in collective choice. In an article accepted at *Public Choice*, also the first chapter of my dissertation, I develop a method for identifying the precise manifestations of un-modeled interdependence in collective choice data. My interest in explicitly modeling the dependencies in political processes lead naturally to an interest in network analysis – an approach ideally situated to dependence modeling. In an article published in *Political Analysis*, my co-author and I review and extend a methodology for modeling and testing hypotheses about specific interdependence processes in network data. Also, in an applied article on international defense pacts accepted at *Conflict Management and Peace Science*, my collaborators and I illustrate how the methods developed in the *Political Analysis* article can be used to incorporate rich qualitative theory about interdependence among states into empirical models of alliance formation.

Models that can be used to infer the dependence among outcomes as well as their dependence upon covariates typically require (1) substantial additional parametric assumptions about model structure and (2) approximate estimation methods with properties that fall short of the methods used for modeling independent observations. These features highlight needs for (a) methods of estimation that are robust to misspecification of the model structure and (b) methods for directly comparing multiple models estimated on the same data, possibly using different estimation techniques (e.g., approximate versus exact maximum likelihood). I have an ongoing research agenda in robust estimation and out-of-sample testing. In an article accepted for publication in *State Politics & Policy Quarterly*, my co-author and I present an alternative to least squares (LS) for estimating the linear model – least absolute deviations (LD) regression, and develop an out-of-sample test for determining whether LD is more appropriate than LS in a given application. Additionally, in an article accepted for publication at *Political Analysis*, my co-author and I present a method for estimating the Cox proportional hazards model that is more robust to model misspecification than the conventional partial likelihood maximization method. We also introduce a test that can be used to determine whether the robust method is more appropriate in a given application. Additionally, I am currently working on methods for count data models.

From my work on out-of-sample testing and dependence modeling, I became interested in methods for leveraging dependence that unfolds dynamically for out-of-sample prediction, i.e., forecasting. In a paper accepted for publication in the proceedings of the IEEE Computer Society, my co-author and I show how machine learning methods for deterministic link prediction for networks can be integrated into a probabilistic framework using temporal exponential random graph models (TERGMs). We apply the link prediction methods to forecasting the attacker origin and location of transnational terrorist attacks, and achieve surprising success. Also, in an article under revise and resubmit at a statistical mechanics journal, *Physica A*, my co-author and I show how link forecasting methods recently developed in the statistical mechanics literature can be integrated with other forecasting approaches using TERGMs. We show that integrated TERGMs out-perform the statistical mechanics methods at forecasting cosponsorship in the U.S. Senate. Overall, dynamic dependence in social and political processes generally, and networks specifically, offers substantial promise for developing models to accurately forecast interactions of many forms. I intend to continue this research agenda.

I have recently begun a substantial project that draws upon all of the aforementioned areas of expertise, which is preliminarily termed the "Government Communication Networks Project". Communication networks, represented primarily by intra-governmental email networks, contain a wealth of information about the structure, performance and trajectory of government processes. Also, due to their electronic nature, email networks can be monitored in real-time to develop actionable predictions and suggestions. Consider, for instance, the debate surrounding the causes of the September 11^{th} terrorist attacks. The 9/11 Report characterized the defense, intelligence and emergency management arms of the federal government as exhibiting far from optimal communication. The argument for the creation of the Department of Homeland Security was, in essence, an argument for improving intra and inter-governmental communication networks. Additionally, email data, in the federal and most state governments, is subject to public information requests. My collaborators in this project and I have begun to collect email data from county governments in North Carolina, and have had data requests fulfilled by more than twenty counties thus far. We are currently working on projects in which we (1) develop methods for identifying communication patterns that depart from the norm (i.e., links that should exist but do not, and those that do exist but should not), and (2) forecast properties of communication network topics in other arenas of government communication (e.g., forecasting what topics migrate from internal communication networks to public meeting agendas). This project has substantial potential to provide scientific insight into the structure of governmental and organizational performance related to communication patterns. It also holds promise to provide actionable, practical tools for governments and other organizations to monitor and improve their communication protocols in near real-time.

Over the next several years I anticipate developing these agendas further, and in many instances, merging them. There is much work to be done in formulating dependence models for social and political networks. For example, the class of exponential random graph models (ERGMs), the most dominant approach to dependence modeling of networks, is only appropriate for modeling relationships measured on a finite scale. However, many political networks do not fit this constraint (e.g., number of bills cosponsored in a legislature, monetary transactions). A co-author and I are working on a generalization of ERGMs to continuous-valued edges. We presented preliminary work on this at the 2011 Political Networks Conference, currently have an article in which we present the derivation of our model under revise and resubmit at *PLoS ONE*, and currently have a major grant proposal under review at the National Science Foundation in which we request support for developing this generalized ERGM. Also, I believe the network analysis community has just begun to scratch the surface on the ways in which dynamic dependence can be leveraged to accurately forecast relational events. This is an exciting area of research that I intend to continue. For example, a co-author and I are currently collecting data on recorded instances of international human trafficking. We will investigate the performance of network-analytic methods in predicting these instances. Lastly, I have high hopes for the Government Communication Networks Project. With the data we are collecting, we will be able to conduct numerous longitudinal and cross-sectional studies of government communication processes. These will lead to substantive findings related to agenda formation, innovation and organizational structure within government. This data also presents much opportunity for methods development. For example, I note above that most network analysis methods are developed for finite or even dichotomous relationship types, and that I am currently working on methods for continuous valued edges. Email networks present an additional challenge. Links are text-valued. Overall, I have a number of well established research agendas that hold potential for substantial additional productivity and innovation moving forward.